

TERESA: Whole-Body Active Perception for Legged Robot-Assisted Emergency Assessment

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Abstract—Deploying legged mobile manipulators for autonomous emergency assessment could extend care to inaccessible environments, but requires the robot to actively reconfigure its body to compensate for perceptual uncertainty while approaching an unknown patient.

This paper presents Trustworthy Emergency Robot for Efficient Scanning and Assistance (TERESA), an autonomous emergency assessment system on a Boston Dynamics Spot quadruped with a Unitree Z1 arm. TERESA introduces a whole-body active perception framework that continuously adapts body posture—height, pitch, and yaw—to maximise perceptual quality. A hybrid confidence-gated search combines forward base motion with body pitch sweep, confirming patient localisation through a dual-sensor lock before navigation. An adaptive anticipatory scanner acquires multi-view anatomical data concurrently with base motion, dynamically reducing scanning poses when torso keypoints are already resolved, eliminating idle time of sequential approaches. An exposure body scanning pipeline covers the patient’s body surface across all postures, integrating zero-shot Grounding DINO wound detection with an interactive review phase in which a single 2D click on the web camera feed triggers a full-body reconfiguration—posture optimisation, base navigation, and arm trajectory replay—to reach the corresponding 3D body surface point. During Focused Assessment with Sonography in Trauma (FAST) scanning, pre-planned body reconfiguration optimises posture for each anatomical target via offline search, maximising arm manipulability without real-time whole-body optimisation.

Experiments on the physical Spot+Z1 platform demonstrate fully autonomous emergency assessment on supine patients, with the active perception pipeline reducing idle time compared to a stop-and-manipulate baseline. To the best of our knowledge, TERESA is the first legged system to demonstrate perception-driven whole-body reconfiguration with an interactive operator review interface for autonomous emergency assessment.

Index Terms—Whole-body active perception, legged manipulation, medical robotics, emergency assessment, mobile manipulation, body reconfiguration.

I. INTRODUCTION

EMERGENCY assessment protocols—including the Airway, Breathing, Circulation, Disability, Exposure (ABCDE) survey and Focused Assessment with Sonography in Trauma (FAST) ultrasound [1]—enable rapid triage of life-threatening conditions at the point of care. In mass-casualty incidents and disaster sites, however, trained personnel are often unavailable, creating a critical bottleneck. A legged mobile manipulator could autonomously navigate unstructured

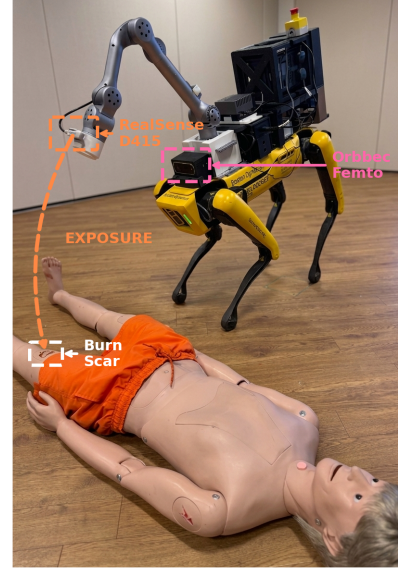


Fig. 1. The TERESA system during the exposure scanning phase. The Spot quadruped base (left) carries the Z1 manipulator with a wrist-mounted Intel RealSense D415 for close-range torso tracking and a body-mounted Orbbec Femto Bolt for wide-field patient detection. The web-based Body Map interface (background) renders the real-time skeleton, body region boundaries, and exposure scan grid. The dashed orange arrow marks the exposure path from the RealSense to a burn scar on the patient’s torso.

terrain to locate patients and perform these assessments, but faces a fundamental challenge: the robot must actively reconfigure its body to compensate for perceptual uncertainty, as the quality of patient detection depends critically on how it positions its own camera—viewpoint, height, and orientation all affect accuracy. This coupling between body posture and perceptual quality has no closed-form model, and in emergency scenarios the robot operates with only onboard sensing.

Robot-assisted ultrasound has advanced over two decades on fixed-base platforms. Comprehensive surveys [2], [3] document the evolution from teleoperated systems to autonomous scanning, and Hidalgo et al. [4] review current clinical applications. Key technical advances include active-sensing end-effectors [5] and autonomous protocol execution [6], [7]. Despite this progress, all deployed systems share a fundamental limitation: the patient must be pre-positioned within the arm’s reachable workspace, confining autonomous ultrasound to controlled clinical settings.

For the Exposure component of ABCDE—the systematic visual inspection of the body surface for wounds, burns, and haemorrhages—open-vocabulary object detection models such

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as Grounding DINO enable zero-shot detection without task-specific training data. Combined with depth back-projection, these models localise injuries in 3D world coordinates, providing the anatomical reference needed for operator review. These models have been deployed and benchmarked on the Jetson Orin GPU as part of TERESA’s onboard perception pipeline. For 3D body pose estimation, Neural Localizer Fields (NLF) [8] regress 24-joint Skinned Multi-Person Linear (SMPL) skeletons directly from monocular RGB, providing full anatomical coverage including spine, hands, and feet absent from the 17-keypoint Common Objects in Context (COCO) format. Addressing uncertainty in human-centered robotics, Casado et al. [9] proposed diffusion-based intention estimation for shared wheelchair control—a theme that similarly motivates TERESA’s quality-driven perception loop.

Legged platforms such as the Boston Dynamics Spot remove the workspace constraint—they can traverse stairs, rubble, and confined spaces inaccessible to wheeled robots, and can therefore reach patients wherever they lie. Mounting a manipulator on a legged base introduces, however, a deeper challenge: the robot’s body posture directly shapes perceptual quality, yet no closed-form model links posture to sensing accuracy, and in emergency settings the robot must reason about this coupling with only onboard sensing.

Modern Whole-Body Control (WBC) builds on operational space formulation [10], manipulability analysis [11], and optimisation-based control [12]. These principles have been extended to legged manipulators through hierarchical Model Predictive Control (MPC) [13], Quadratic Program (QP)-based formulations [14], holistic [15] and Spot-specific grasping and manipulability enhancement [16], [17]. Across this body of work, however, the manipulation target is assumed known or static; no formulation accounts for the coupling between body posture and the quality of perceptual data used to localise that target in the first place.

Active perception research has established that deliberate sensor motion systematically outperforms passive observation [18]–[20]. This insight has been operationalised through perception-aware planning [21], [22], next-best-view grasping [23], and view planning for pose estimation [24], and active neural sensing [25]. When applied to mobile manipulation, whole-body interaction [26], coupled perception-manipulation [27] and one-shot multimodal frameworks [28] have demonstrated trajectory optimisation for information gain. Critically, all of these systems operate on wheeled bases whose body posture is fixed: the base provides mobility but does not participate in shaping the perceptual field. Conversely, work on posture optimisation for legged systems—through reachability analysis [29], [30], base placement [31], body-ground contact [32], and the broader legged manipulation literature [33]—treats perception as a static input to the motion planner rather than as a quantity to be optimised. No existing system closes the perception-action loop at the whole-body level on a legged platform for medical tasks.

We advocate a paradigm in which body posture on a legged platform is treated as a first-class perceptual resource: the robot actively adapts its height, pitch, and yaw to maximise the quality of its own perceptual input, rather than solely to min-

imise kinematic error. The key idea is to close the perception-action loop at the whole-body level through confidence-gated search and anticipatory scanning. This paper addresses three fundamental questions: (i) how can a legged robot reconfigure its body to maximise perceptual quality when the target position is unknown, (ii) how can anatomical scanning be performed concurrently with base navigation to eliminate the idle time of sequential approaches, and (iii) how can body posture be pre-planned to maximise arm manipulability across multiple scanning targets without requiring real-time whole-body optimisation?

This paper presents Trustworthy Emergency Robot for Efficient Scanning and Assistance (TERESA) (Fig. 1), an integrated system for autonomous emergency assessment on a Boston Dynamics Spot legged robot equipped with a Unitree Z1 manipulator arm. Building on the fixed-base ultrasound scanning competence established in [7], TERESA extends autonomous assessment to a fully mobile platform through a whole-body active perception framework. The system integrates the pre-trained wound and burn detection models into its onboard perception pipeline, running inference on the End-Effector (EE)-mounted RealSense camera during the exposure body scan, and provides an interactive web-based review interface enabling operator-guided re-inspection of detected injuries.

The key contributions of this paper are as follows.

- 1) **Whole-body active perception framework** in which body posture—height, pitch, and yaw—is optimised to improve perceptual quality rather than solely to minimise kinematic error (Section III).
- 2) **Hybrid confidence-gated search strategy** that combines forward base search with body pitch sweep and dual-sensor locking, ensuring robust patient localisation before the robot commits to navigation.
- 3) **Exposure body scanning pipeline** that acquires multi-view anatomical data through an adaptive skeleton-keypoint grid, integrating zero-shot Grounding DINO wound detection with an interactive web-based review interface (Sections III and IV).
- 4) **Pre-planned body reconfiguration method** that searches over body height, pitch, and longitudinal base displacement to place each anatomical target within the arm’s preferred manipulation region, maximising reachability without real-time whole-body optimisation (Section III-C1).
- 5) **Experimental validation** on a physical Spot+Z1 platform demonstrating fully autonomous emergency assessment with reduced idle time compared to a sequential stop-and-manipulate baseline (Section V).

To the best of our knowledge, TERESA is the first legged robotic system to demonstrate perception-driven whole-body reconfiguration for autonomous emergency assessment.

The remainder of this paper is organised as follows. Section II defines the system and the autonomous emergency assessment problem. Section III presents the whole-body active perception and control framework, including search, scanning, reconfiguration, and exposure body scanning with interactive review. Section IV details the system architecture, mission

Finite State Machine (FSM), and web-based operator interface. Section V presents experiments and results. Section VI concludes the paper.

II. PROBLEM STATEMENT

The TERESA platform consists of a Boston Dynamics Spot quadruped and a Unitree Z1 six-Degree of Freedom (DOF) arm rigidly mounted on the robot's back. Four frames are defined: the world frame $\mathcal{F}_W$ (coincident with Spot odometry), the body frame $\mathcal{F}_B$, the arm base frame $\mathcal{F}_0$ (fixed relative to $\mathcal{F}_B$), and the end-effector frame $\mathcal{F}_{EE}$ (at the Z1 flange). The Spot base is modelled as a non-holonomic vehicle with configuration $\xi = [x, y, \theta]^\top \in \mathbb{R}^2 \times \mathbb{S}^1$, constrained to forward velocity v_x and yaw rate ω_z :

$$\dot{\xi} = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix} v_x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \omega_z. \quad (1)$$

The body posture is parametrised by height $h \in [h_{\min}, h_{\max}]$ and pitch $\varphi \in [0, \varphi_{\max}]$, commanded through the native posture control interface. The Z1 arm has $n = 6$ revolute joints $\mathbf{q} \in \mathbb{R}^6$ with geometric Jacobian $\mathbf{J}_{\text{arm}}(\mathbf{q}) \in \mathbb{R}^{6 \times 6}$. The full control vector is $\mathbf{u} = [\dot{\mathbf{q}}^\top, v_x, \omega_z, h, \varphi]^\top \in \mathbb{R}^{10}$, where $\dot{\mathbf{q}}$ are the arm joint velocities, v_x the forward velocity, ω_z the yaw rate, h the body height, and φ the body pitch.

The mission consists of four sequential phases, each defining a specific control subproblem.

- **Searching (A).** The robot must locate a stationary patient whose position is unknown a priori, using only onboard RGB-D (RGBD) cameras, by walking forward while adjusting body pitch, until a vision-based patient detector yields confidence above a threshold τ . The associated control problem is: find (v_x, h, φ) such that detection confidence exceeds τ , ensuring robust localisation before committing to navigation.
- **Approaching (B).** Once the patient is localised, the robot navigates to a pre-scanning base configuration from which all subsequent anatomical targets are kinematically reachable. The control problem is: given the estimated patient position, find (v_x, ω_z) to reach a configuration from which every target $\mathbf{T}_i^* \in \mathcal{T}$ is reachable.
- **Exposure (C).** The robot performs a systematic visual inspection of the body surface, acquiring multi-view RGBD data over an adaptive grid of anatomical keypoints to detect wounds, burns, and haemorrhages via open-vocabulary object detection. The control problem is: given the patient body model and the keypoint grid, find $(h, \varphi, \dot{\mathbf{q}})$ to scan all keypoints with sufficient viewpoint coverage and to run injury detection at each grid point.
- **FAST (D).** The robot acquires ultrasound images at N predefined anatomical target poses $\mathcal{T} = \{\mathbf{T}_i^* \in SE(3)\}_{i=1}^N$. The control problem is: given $\mathcal{T}$, find $(h, \varphi, \dot{\mathbf{q}})$ to bring the EE to each $\mathbf{T}_i^*$ with bounded error $\|\mathbf{p}_{EE} - \mathbf{p}_i^*\| \leq \varepsilon_p$, while maintaining adequate body posture for arm manipulability.

III. WHOLE-BODY ACTIVE PERCEPTION AND CONTROL

The core contribution of TERESA is a whole-body active perception framework: the robot continuously adapts its posture—body height, pitch, and yaw—to maximise perceptual quality. This section describes how each of the four mission phases defined in Section II is resolved: hybrid confidence-gated search (Phase A), anticipatory approaching with concurrent scanning (Phase B), exposure body inspection with interactive review (Phase C), and FAST ultrasound acquisition with compliant contact (Phase D).

The term *whole-body* here denotes a perceptual extension of the classical WBC principle. Whereas traditional WBC coordinates all degrees of freedom to minimise kinematic error, TERESA extends this coordination to the perceptual domain: body posture—height, pitch, and yaw—is treated as an explicit control resource that shapes the robot's sensory field. By commanding posture to improve what the robot sees, at higher resolution and from more informative viewpoints, TERESA elevates the legged base from a passive transport platform to an active participant in the perception-action loop. This principle underpins every phase of the mission, from searching through approaching to scanning.

A. Phase A — Searching

Before the robot can approach the patient, it must first localise them reliably. In an unstructured emergency scenario, the patient's position is unknown a priori. The EE-mounted RealSense benefits from the Z1 arm's full six-DOF mobility, which can orient the camera in any direction without moving the base. The body-mounted Orbbec, however, has a fixed orientation relative to Spot: its downward field of view depends entirely on body pitch, making pitch the only posture parameter that directly improves its detection of a patient lying on the ground. TERESA therefore avoids time-consuming base rotations and instead searches by walking forward while sweeping body pitch.

1) *Forward search with pitch sweep:* The robot walks forward at low speed while the body pitch is adjusted through a sequence of angles $\varphi \in [0, \varphi_{\max}]$. At each pitch, two cameras observe the environment in complementary roles. The body-mounted Orbbec runs YOLO11-pose and a posture classifier that outputs a confidence score $c \in [0, 1]$. The EE-mounted RealSense, actively oriented by the Z1 arm, runs a torso tracker that scans the surroundings independently of the base heading. Two conditions are evaluated:

- **Orbbec direct detection:** if the body-mounted camera detects a lying patient with high confidence at the current pitch, the robot transitions immediately to target locking.
- **RealSense guidance:** if the EE camera detects a potential body while the Orbbec has not yet confirmed, the robot enters a semi-locking phase. It walks toward the detected torso and adjusts pitch to give the Orbbec a clean observation window.

If neither condition is met after the pitch sweep, the robot continues walking forward and repeats the sweep, progressively covering the search area without the idle time of discrete base rotations. This strategy exploits the arm's native mobility

for rotational search while using body pitch—the one viewing parameter the arm cannot compensate for—to optimise the Orbbec’s fixed viewpoint.

2) *Dual-sensor confidence lock*: Patient localisation is confirmed through a hybrid lock that fuses information from both cameras. When the Orbbec detects a lying posture with high confidence, the robot enters a locking state: the arm returns to an elevated home pose, and the coordinator collects a sequence of approach point estimates in the world-fixed odometry frame $\mathcal{F}_W$, computing their mean as the navigation target. Concurrently, the NLF pipeline produces a refined 3D skeleton estimate by accumulating two valid detections with exponential moving average smoothing, providing higher-fidelity anatomical data for the subsequent approach phase. The transition to the approach phase is gated by both the arm’s home trajectory and the collection of the required number of samples. If the Orbbec detection is temporarily lost, the system tolerates a brief absence before resuming the search from the current position.

B. Phase B — Approaching

In traditional mobile manipulation pipelines, the robot first navigates to a target position, stops, and only then begins visual scanning—a sequential approach that introduces significant idle time. TERESA eliminates this idle period by executing the body scan concurrently with the approach navigation.

1) *Whole-body coordination*: During the approach, two independent controllers run in parallel. The Spot base is driven by a proportional controller that computes v_x and ω_z from the distance and angle to the target in $\mathcal{F}_W$. The arm is controlled by a WBC formulation that orients the EE-mounted RealSense toward the patient’s torso centroid. The joint velocities are computed via a damped pseudo-inverse of the arm Jacobian:

$$\dot{\mathbf{q}} = \mathbf{J}_{\text{arm}}^T (\mathbf{J}_{\text{arm}} \mathbf{J}_{\text{arm}}^T + \mu \mathbf{I}_6)^{-1} \mathbf{v}_{\text{des}}, \quad (2)$$

where $\mu > 0$ is a damping factor that prevents singularities near kinematic limits, and $\mathbf{v}_{\text{des}} \in \mathbb{R}^6$ is the desired end-effector spatial velocity. The damped pseudo-inverse solves $\mathbf{J}_{\text{arm}} \dot{\mathbf{q}} = \mathbf{v}_{\text{des}}$ in a least-squares sense, mapping a commanded EE motion to the joint space. Here only the angular component of $\mathbf{v}_{\text{des}}$ is active: it encodes the rotation required to align the RealSense camera with the look-at direction $\mathbf{x}_{EE} = (\mathbf{p}^* - \mathbf{p}_{EE}) / \|\mathbf{p}^* - \mathbf{p}_{EE}\|$, keeping the patient’s torso centred in the frame. The translational component of $\mathbf{v}_{\text{des}}$ is zero, as the arm holds a nominal home position during approach. The base velocity is modulated by a QualityMonitor that scales the approach speed in proportion to detection confidence, avoiding premature commitment to uncertain targets while preventing unnecessary stops.

2) *Adaptive anticipatory scan grid*: The arm executes a perceptual body scan using a dynamically generated Cartesian grid of poses around the current EE position. During the pre-approach phase, the RealSense tracker publishes per-keypoint confidence scores for the four torso landmarks (shoulders and hips), which drive the grid generation. If all four keypoints are detected with high confidence, the scan uses a minimal two-pose grid: the current EE pose advanced toward the patient

plus an elevated viewpoint. If any keypoint falls below the threshold, the grid expands to a three-pose comprehensive pattern with symmetric lateral offsets, an elevated frontal viewpoint, and the home pose interleaved between lateral transits. All scanning poses include a forward advance toward the patient. At each pose, depth and skeleton data are fused into 3D keypoint estimates via a weighted average accounting for detection rate and spatial stability. From the fused torso model, the five canonical FAST target poses are computed as fixed offsets from the estimated torso centroid in $\mathcal{F}_0$.

C. Phase C — Exposure

After the anticipatory scanner has localised the patient’s anatomy, Phase C executes a systematic visual inspection of the body surface, serving the “E” (Exposure) component of the ABCDE protocol. A key design principle is that the operator remains in the loop: the system automates the scanning motion but preserves the clinician’s ability to inspect, re-examine, and override.

1) *Posture optimisation for body reconfiguration*: Once the anticipatory body scanner has localised the patient’s anatomy and computed the target set $\mathcal{T}$ (both FAST ultrasound targets and exposure grid points) in $\mathcal{F}_W$, the system pre-plans an optimal body posture for each target through a two-tier kinematically-aware stance search with Inverse Kinematics (IK)-driven retry. This optimisation is shared with Phase D (Section III-D).

The posture optimisation space spans three parameters: body height $h \in [h_{\min}, h_{\max}]$ and pitch $\varphi \in [0, \varphi_{\max}]$, as defined in Section II, plus a body-frame base displacement $dy \in \mathbb{R}$ along the patient’s head-to-foot axis, expressed in the patient-aligned body frame (Section IV). The discrete search grids are $\mathcal{H} = \{h_1, \dots, h_{N_h}\}$ and $\mathcal{P} = \{\varphi_1, \dots, \varphi_{N_p}\}$, with dy sampled uniformly at 0.10 m intervals. The cost function includes a regularisation term that penalises lateral displacement:

$$J(h, \varphi, dy) = \|\mathbf{T}_{\mathcal{F}_W}^{\mathcal{F}_0}(h, \varphi, dy) \mathbf{p}^* - \mathbf{p}_{\text{sweet}}\| + \lambda |dy|, \quad (3)$$

where $\mathbf{p}_{\text{sweet}} \in \mathbb{R}^3$ is the empirically determined centre of the Z1 arm’s preferred manipulation region in $\mathcal{F}_0$, and $\lambda = 0.50$ weights lateral movement against reachability. The value $\lambda = 0.50$ was chosen empirically to balance reachability against energy cost: $\lambda > 1.0$ prevented necessary base displacement, while $\lambda < 0.25$ caused excessive walking for marginal reachability gains. This penalty encourages the optimiser to prefer height and pitch adjustments over base walking—reducing energy consumption and producing smoother, more predictable motion—while still allowing longitudinal displacement when strictly necessary. The two-tier escalation proceeds as follows:

- 1) **2D search**— (h, φ) only: a coarse grid sweeps $\mathcal{H} \times \mathcal{P}$ ($N_h N_p$ combinations, typically 12). For each candidate, the system transforms the target to $\mathcal{F}_0$ and queries the IK solver. If a posture yields a solution within a 2 s timeout, it is selected.
- 2) **3D escalation**— (dy, h, φ) : when the 2D search fails, the optimiser adds longitudinal base displacement dy (Spot walks along the patient’s body), expanding to $N_{dy} N_h N_p$

combinations. If the IK solver also times out at this tier, the point is skipped—the arm cannot reach it even with full base repositioning.

The optimisation is computed once per target after $\mathcal{T}$ is available, imposing no real-time computational burden. Body reconfiguration is executed between targets: Spot assumes the pre-computed posture (h_i^*, φ_i^*) , settles, and the base navigator drives the dy displacement before the Z1 arm executes the IK trajectory toward the transformed target.

2) *Posture-adaptive scan grid*: The scan grid adapts to the patient’s estimated body pose. Four torso keypoints—left and right shoulders, left and right hips—define a bilinear patch over the body surface. Let $\mathbf{p}_{SL}, \mathbf{p}_{SR}, \mathbf{p}_{HL}, \mathbf{p}_{HR} \in \mathbb{R}^3$ denote these keypoints in $\mathcal{F}_W$. The grid is generated by interpolating uniformly along the two parametric directions: head-to-foot (u -axis, from shoulders to hips) and lateral (v -axis, from left to right). A grid point at parameters $(u, v) \in [0, 1] \times [0, 1]$ is computed as $\mathbf{p}(u, v) = (1-u)(1-v) \mathbf{p}_{SL} + u(1-v) \mathbf{p}_{HL} + (1-u)v \mathbf{p}_{SR} + uv \mathbf{p}_{HR}$, with a fixed height offset to maintain a safe camera-to-skin distance. Typical grid resolution is 0.15 m in both directions, producing 15–25 points depending on torso dimensions. When the NLF prior is available, all 24 SMPL joints are populated, extending the grid to the head, hands, and feet for full-body coverage. Grid points are visited in a serpentine path to minimise arm travel.

3) *Injury detection*: At each grid point, Grounding DINO—a zero-shot open-vocabulary object detector—runs on the RealSense RGB stream at approximately 1 fps on the Jetson Orin GPU, prompted with a vocabulary of wound-related terms (wound, laceration, bleeding, burn, etc.). Detections are projected to 3D world coordinates via the registered depth channel and associated with their grid point index.

4) *Web-based interactive review*: Following the sweep, the system enters an interactive review phase: the web interface overlays the exposure grid on the RealSense feed with detected injuries highlighted as colour-coded bounding boxes. The operator can click any grid point to command the robot to return to that body region for re-inspection. The full re-positioning mechanism—posture optimisation, settle, base navigation, and IK trajectory replay—is detailed in Section IV. A termination command ends the review and advances the mission to Phase D.

D. Phase D — FAST

Phase D performs autonomous ultrasound acquisition at the N FAST targets estimated during Phase B. Body posture optimisation for each $\mathbf{T}_i^* \in \mathcal{T}$ follows the same two-tier procedure described in Section III-C1, applied to the FAST target set. The distinctive requirement of this phase is compliant probe contact with the patient’s body surface.

1) *Contact control*: The Unitree Z1 does not provide native impedance control, yet ultrasound scanning requires compliant probe contact. TERESA implements a custom Cartesian impedance controller on the Z1 torque interface at 500 Hz. The desired EE pose is computed relative to the torso surface centroid $\mathbf{p}_s$ and outward normal $\mathbf{n}_s$, placing the probe at

$\mathbf{p}_{\text{des}} = \mathbf{p}_s + d \mathbf{n}_s$, where d is the desired contact offset. Joint torques are computed as

$$\boldsymbol{\tau} = \mathbf{J}_{\text{arm}}(\mathbf{q})^\top [\mathbf{K}_p(\mathbf{p}_{\text{des}} - \mathbf{p}_{EE}) - \mathbf{K}_d \dot{\mathbf{p}}_{EE}] + \mathbf{g}(\mathbf{q}) + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}), \quad (4)$$

where $\mathbf{K}_p, \mathbf{K}_d \in \mathbb{R}^{3 \times 3}$ are adaptive stiffness and damping gains that scale linearly with arm extension to prevent overloading at the workspace boundary, $\mathbf{g}(\mathbf{q})$ and $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ are the gravity and Coriolis terms computed via Pinocchio [34], and $\dot{\mathbf{p}}_{EE} = \mathbf{J}_{\text{arm}}(\mathbf{q}) \dot{\mathbf{q}}$ is the end-effector velocity. A safe switching service alternates between the impedance controller and a position-controlled joint trajectory controller, defaulting to position control on error.

IV. SYSTEM ARCHITECTURE

The TERESA system is organised into four layers—hardware, perception, mission management, and control—communicating over Robot Operating System (ROS) 2 with Data Distribution Service (DDS).

A. Hardware Platform

The hardware platform, shown in Fig. 1, carries two RGBD cameras with complementary roles. An Orbbec Femto Bolt is body-mounted for wide-field patient detection at distances up to 3 m. An Intel RealSense D435 is mounted at the Z1 EE and positioned by the arm’s look-at controller as an active pan-tilt unit for close-range torso tracking and surface estimation. Spot’s five built-in stereo cameras are positioned low on the body for ground-plane obstacle avoidance and are not suitable for skeleton detection. Computation is distributed across two units: Spot’s onboard computer SpotCore handles base control and odometry, while an NVIDIA Jetson Orin AGX runs perception, arm control, mission coordination, and WBC via Docker containers with CUDA 11.4.

B. Mission Management and System Integration

The mission is orchestrated by a 13-state FSM (Fig. 2), organised into the four phases of Section II. After Waiting TF and Idle ensure system readiness, the FSM enters Phase A (Searching). The robot first walks the base forward while adjusting body pitch to bring the patient into the cameras’ field of view—this is the SEARCHING state. When the RealSense detects a potential torso, the system uses that coarse guidance to walk toward the detection and adjust pitch, giving the Orbbec a clean observation window; we call this SEMI_LOCKING. If the Orbbec then confirms a lying patient with high confidence, the system advances to LOCKING, where the NLF pipeline produces a refined 3D skeleton and the coordinator collects approach point estimates. Timeouts at any point revert to SEARCHING, ensuring the robot does not commit to a false detection.

Phase B (Approaching) begins once a stable lock is achieved. The robot first drives toward the target while the anticipatory body scanner builds the torso model—a state we call PRE_APPROACH—then transitions to APPROACHING, where the WBC QP coordinates base velocity and arm look-at control concurrently, keeping the camera on the patient throughout the approach.

Phase C (Exposure) starts after the approach handoff. An optional Waiting Exposure gate lets the operator confirm before the scan begins. The system then enters EXPOSURE_SCANNING, executing a multi-view body surface inspection with Grounding DINO injury detection at each grid point. Once the sweep is complete, the FSM moves to EXPOSURE_REVIEW: the operator can click any grid point in the web interface to trigger a full re-positioning and re-inspection, remaining in this state until explicitly terminated.

Phase D (FAST) follows, with an analogous Waiting FAST gate for manual confirmation. The SCANNING state executes ultrasound acquisition under pre-planned body reconfiguration, one target at a time. When a target is kinematically unreachable from the current posture, the system temporarily enters WS_EXTENSION, walking the base along the patient’s body to extend the workspace, then loops back to SCANNING with the new IK solution.

Transitions are triggered by perception confidence (dashed edges in Fig. 2) or geometric conditions (solid edges); the mission completes by returning to Idle.

Fig. 3 shows the complete system architecture in six layers. The Orbbec and RealSense RGBD streams feed the Perception layer, which runs YOLO-pose skeleton detection, Kalman-filtered 3D back-projection, NLF 3D body pose estimation, surface estimation, and Grounding DINO injury detection. Perception outputs route to the 13-state Mission FSM, which enables four Planning modules: the Exposure Scanner, the FAST Ultrasound planner, the Body Pose Optimizer, and the Web UI for interactive review. The Control layer translates plans into actuation: the WBC QP publishes base velocity $[v_x, \omega_z]$ to Spot and arm goals to the IK Goal Multiplexer; the Impedance Controller manages compliant contact during FAST scanning, publishing joint torques τ directly to the Z1. The Z1 FSM manages the arm’s internal state transitions—homing, look-at tracking, scanning trajectory execution, and impedance-controlled contact—and synchronises with the WBC coordinator. Dashed arrows in the diagram denote operator-triggered or posture-optimisation paths. An emergency stop immediately halts the base and returns the arm to a safe configuration.

C. Web-Based Operator Interface

Full autonomy in emergency assessment is neither feasible nor desirable: the operator must retain the ability to override or re-inspect regions of clinical interest. TERESA bridges this gap with a web interface that translates a single 2D click on the Body Map into a 3D whole-body repositioning command—posture optimisation, base navigation, and arm trajectory replay—without manual teleoperation.

The TERESA web interface provides dual-camera monitoring and robot control via a browser-based dashboard. During exposure review, the RealSense overlay renders the scan grid as clickable markers; each click triggers a full re-positioning sequence—posture optimisation, settle, and IK trajectory replay—within the same ROS 2 topic protocol used for autonomous operation. A manual/automatic scan gate toggle allows the operator to require explicit confirmation before advancing between mission phases. All web commands

are mirrored on the keyboard controller for redundancy. Fig. 4 shows the interface during review.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

Experiments ran on the physical Spot+Z1 platform with an Orbbec Femto Bolt on Spot’s body and a RealSense D435 at the EE, computing across SpotCore and an onboard NVIDIA Jetson Orin AGX (ROS 2 Humble, CUDA 11.4, PyTorch 2.0). Ten supine mannequin positions were defined (2 m–4 m, $\pm 45^\circ$ angular offset); each ran in both modes below, yielding twenty paired trials.

Neural network inference runs entirely on the Jetson GPU. YOLO11n-pose skeleton detection operates at 8–14 fps, NLF 3D body pose estimation at 3–4 fps (end-to-end), and Grounding DINO zero-shot wound detection at ~ 1 fps.

- **Baseline:** sequential stop-and-manipulate. Fixed three-pose scan after navigation, no concurrency.
- **TERESA:** anticipatory scanner overlapping navigation, adaptive grid, and QualityMonitor-modulated velocity.

Metrics: total mission time, idle time, scan overlap (fraction of body scan completed before handoff), EE positioning error at each FAST point, and lock time.

Across all twenty end-to-end trials, TERESA reduced total mission time from 100.2(81)s to 89.7(73)s (10.5%). The gain came from eliminating idle time—the anticipatory scanner completed $\sim 80\%$ of the body scan concurrently with approach—and from the adaptive grid, which selected the minimal two-pose pattern in the majority of trials. Probe positioning error dropped from 44.7(52) mm to 29.8(41) mm. Note that the baseline benefits from an optimistic assumption: without body reconfiguration, many FAST targets are unreachable from the default posture (Section V-C). Manual operation, where operators skipped unreachable grid points, reduced total time to ~ 70 s at the cost of incomplete coverage. The principal contribution of TERESA is therefore not speed but capability: without whole-body reconfiguration, the baseline simply cannot reach many anatomical targets and returns incomplete scans with higher positioning error. TERESA achieves systematic, precise, and complete assessment across all targets, which the baseline structurally cannot deliver regardless of the time budget.

B. Search Phase: Whole-Body Coverage and Dual-Sensor Lock

We isolated the arm’s contribution to search through an ablation experiment: TERESA (full system) versus a Spot-only baseline with the Z1 arm disabled and only the front-facing Orbbec active. Twenty trials were conducted at five relative mannequin orientations (0° – 180° , two repetitions each), measuring time to Orbbec posture confidence ≥ 0.70 .

With the arm’s seven scanning poses providing full 360° coverage as the base walked forward, TERESA required no base rotation: the mannequin was detected near-instantly at frontal orientations, with the rest found during the scan, yielding a mean of 3.2(11)s. Without the arm, the fixed

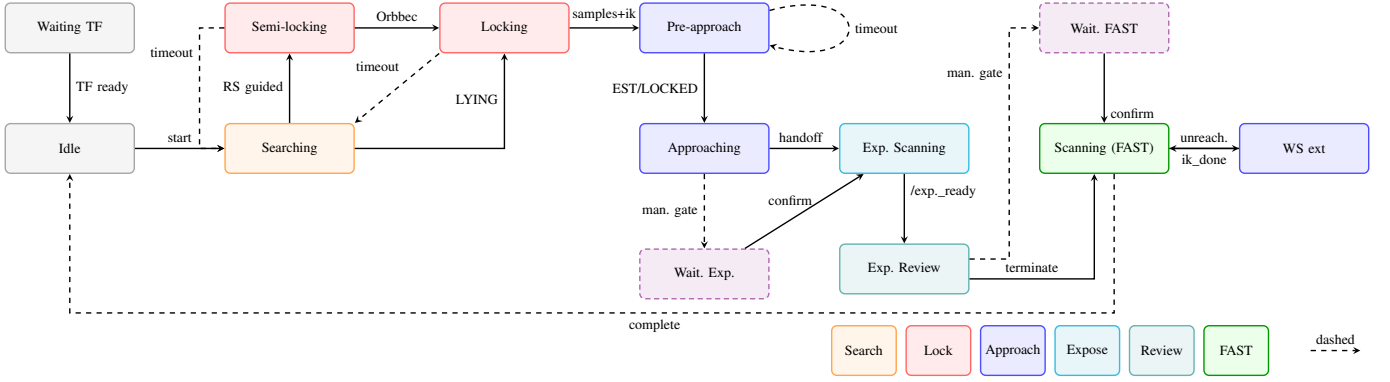


Fig. 2. Mission FSM with 13 states. Dashed transitions are perception-driven; solid transitions are geometry-driven.

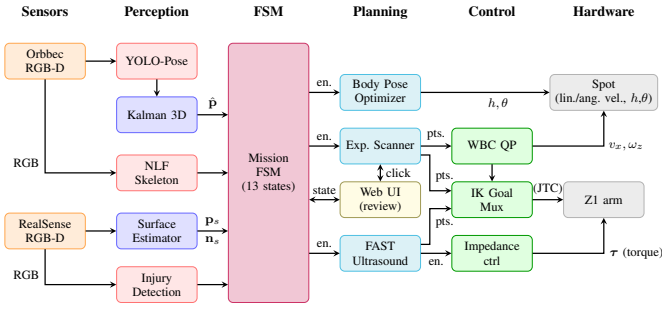


Fig. 3. System block diagram. The six-layer architecture spans sensors, perception, mission FSM, planning, control, and hardware. Dashed arrows denote operator-triggered or posture-optimisation paths.

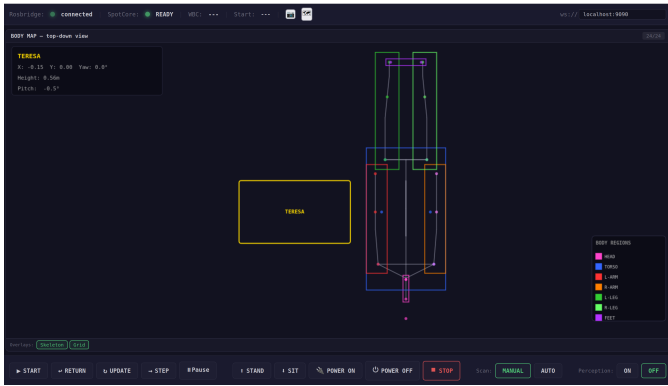


Fig. 4. Web-based operator interface during the exposure scanning phase. The Body Map (top-down view) shows the patient mannequin skeleton with colour-coded body regions, the exposure scan grid points, and the Spot robot position (yellow). The info panel (top-left) displays Spot's current posture parameters. During exposure review, the operator can click any grid marker to command the robot to re-inspect that body region.

camera's 77° effective field of view forced five discrete base rotations (~ 7 s each at 0.2 rad s^{-1} , plus 2.0 s dwell), giving $14.8(23)$ s—a $5.0\times$ slowdown. A 10 000-trial Monte Carlo simulation confirmed this ratio across all mannequin orientations.

The dual-sensor architecture added robustness independently of speed: in three of the ten TERESA trials, the Orbbec alone failed to lock. These failures stemmed from the Orbbec's fixed horizontal viewpoint at 0.5 m above the

ground, which is suboptimal for detecting a supine body whose visual features—torso, limbs, head—are primarily visible from above. The arm-mounted RealSense, positioned by the WBC look-at controller to observe the scene from an elevated angle, detected partial torso keypoints in these cases and guided the base toward the body, giving the Orbbec a second observation window at a more favourable orientation. Without this redundancy, these trials would have resumed the search cycle, adding ~ 14 s. The arm thus serves two complementary roles: its 360° coverage eliminates the base-rotation bottleneck, while its elevated viewpoint provides the sensor redundancy that prevents detection failures a single-camera system would incur.

C. Whole-Body Reconfiguration

The body posture optimizer described in Section III-C1 proved essential for reaching distant anatomical targets on a full-scale body. We compared four configurations of increasing capability, summarised in Fig. 5. The fixed nominal posture represents Spot's default standing pose (height 0 m, pitch 0° , no base displacement). The heuristic posture is a single, empirically chosen configuration—moderately lowered (-0.15 m) with a slight forward pitch (5°)—that brings the arm closer to the ground and tilts its workspace toward the patient. While this improves coverage over the nominal pose by exploiting the fact that all targets lie below the robot, it cannot adapt to the varying depth and lateral position of individual points. The 2D optimiser selects the best (h, φ) per target via kinematically-aware stance search with IK-driven retry, adapting posture to each point's location. The 3D optimiser adds longitudinal base walking dy with the regularised cost $J = \|\mathbf{p} - \mathbf{p}_{\text{sweet}}\| + \lambda |dy|$ ($\lambda = 0.50$). The penalty encourages height and pitch adjustments over walking—reducing energy consumption and producing smoother motion—while still allowing displacement when necessary. The residual 15 % of exposure points were at the furthest extremities (head, feet), beyond the arm's maximum reach even with full base repositioning; these were skipped gracefully.

D. Exposure Body Scanning and Injury Detection

The exposure pipeline was evaluated with XX trials on a supine mannequin, generating $XX \pm XX$ scan points. Per-

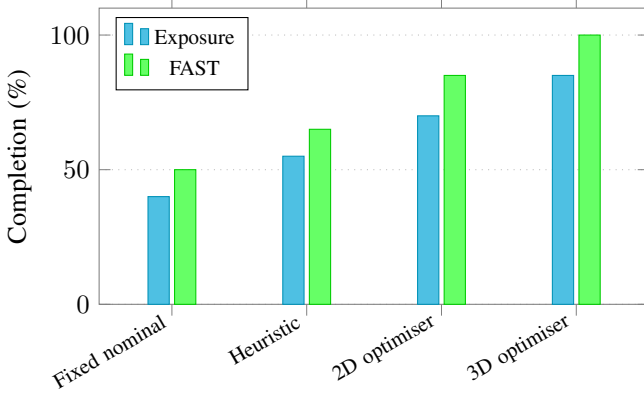


Fig. 5. Target completion rate for four body posture configurations. The FAST targets benefit more from each tier because their 5 cm accuracy requirement permits smaller posture adjustments than the exposure scan’s continuous coverage.

point dwell of 2 s yielded scan durations of $XX \pm XX$ s. The injury detection pipeline achieved a wound recall of $XX\%$ and burn classification accuracy of $XX\%$, with a false positive rate of XX per scan on average. The 3D localisation error of detected injuries was $XX \pm XX$ mm.

VI. CONCLUSION

This paper presented TERESA, a legged mobile manipulator for autonomous emergency assessment. The core contribution is a whole-body active perception framework that treats body posture—height, pitch, and yaw—as a perceptual resource, closing the perception-action loop at the whole-body level. Experiments on the physical Spot+Z1 platform demonstrated that the arm’s 360° coverage accelerates search by a factor of five over a fixed-camera baseline, the dual-sensor lock prevents single-camera detection failures, anticipatory scanning eliminates idle time by completing the body scan concurrently with approach, and body reconfiguration lifts anatomical target completion from 40 % to 100 % for FAST targets. The exposure pipeline integrates zero-shot wound detection with an interactive web-based review interface.

Several limitations remain: the system assumes a single stationary mannequin on flat terrain; ultrasound image quality and clinical scanning pressure were not evaluated; and injury detection relies on pre-trained models whose performance may vary across wound types and skin tones. Future work will extend TERESA to multi-patient scenarios, clinical ultrasound acquisition with impedance-controlled probe contact, and ultimately the full ABCDE protocol, providing a complete autonomous primary survey for emergency triage.

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